

1. Write a Pandas program to create a Pivot table and find the item wise unit sold. .(refer sales_data table)

Code:

```
import pandas as pd

# Sales Data
data = {
    'Item': ['Television', 'Home Theater', 'Television', 'Cell Phone',
            'Television', 'Home Theater', 'Television', 'Television',
            'Television', 'Home Theater', 'Television', 'Home Theater',
            'Home Theater', 'Television', 'Desk', 'Video Games',
            'Home Theater', 'Cell Phone'],
    'Units': [95, 50, 36, 27, 56, 60, 75, 90, 32,
             60, 90, 29, 81, 35, 2, 16, 28, 64]
}

df = pd.DataFrame(data)

# Pivot Table
pivot_table = pd.pivot_table(
    df,
    values='Units',
    index='Item',
    aggfunc='sum'
)

print(pivot_table)
```

Output:

Item	Units
Cell Phone	91
Desk	2
Home Theater	308
Television	509
Video Games	16

2. Write a Pandas program to create a Pivot table and find the total sale amount region wise, manager wise, sales man wise. .(refer sales_data table)

Code:

```
import pandas as pd

data = {
    'Item': ['Television', 'Home Theater', 'Television', 'Cell Phone',
            'Television', 'Home Theater', 'Television', 'Television',
            'Television', 'Home Theater', 'Television', 'Home Theater',
            'Home Theater', 'Television', 'Desk', 'Video Games',
            'Home Theater', 'Cell Phone'],

    'Sale_amt': [113810, 25000, 43128, 6075, 67088, 30000,
                 89850, 107820, 38336, 30000, 107820,
                 14500, 40500, 41930, 250, 936,
                 14000, 14400]
}

df = pd.DataFrame(data)

# Pivot Table
pivot_table = pd.pivot_table(
    df,
    values='Sale_amt',
    index='Item',
    aggfunc=['max', 'min']
)

print(pivot_table)
```

Output:

	max	min
Item	Sale_amt	Sale_amt
Cell Phone	14400	6075
Desk	250	250
Home Theater	40500	14000
Television	113810	38336
Video Games	936	936

2. Create a dataframe of ten rows, four columns with random values. Write a Pandas program to highlight the negative numbers red and positive numbers black.

Code:

```
import pandas as pd
import numpy as np

# Create DataFrame with 10 rows and 4 columns
np.random.seed(10)

df = pd.DataFrame(
    np.random.randint(-20, 21, size=(10, 4)),
    columns=['A', 'B', 'C', 'D']
)

print("Original DataFrame:")
print(df)

# Function to highlight values
def highlight_color(val):
    if val < 0:
        return 'background-color:red; color:white'
    else:
        return 'background-color:black; color:white'

# Apply styling (Pandas 2.1+)
styled_df = df.style.map(highlight_color)
```

```
# Display styled DataFrame
```

```
print(styled_df)
```

Output:

	A	B	C	D	E
0	1	1.32921	-0.770033	-0.31628	-0.99081
1	2	-1.07082	-1.43871	0.564417	0.295722
2	3	-1.6264	0.219565	0.678805	1.88927
3	4	0.961538	0.104011	-0.481165	0.850229
4	5	1.45342	1.05774	0.165562	0.515018
5	6	-1.33694	0.562861	1.39285	-0.063328
6	7	0.121668	1.2076	-0.00204021	1.6278
7	8	0.354493	1.03753	-0.385684	0.519818
8	9	1.68658	-1.32596	1.42898	-2.08935
9	10	-0.12982	0.631523	-0.586538	0.29072

